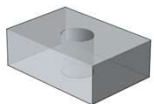
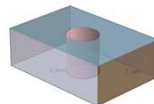


B-rep 输入

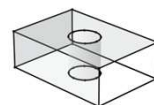
Brep



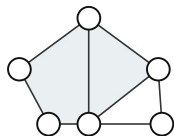
Face



Edge



Topology



Point-level NURBS Serialization

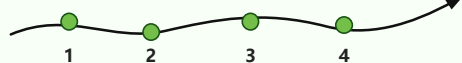
A) Face Sampling

每个 NURBS 曲面在 UV 参数域得到进行 4×4 均匀采样, 得到 16 个三维点



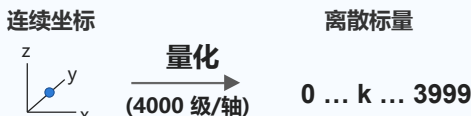
B) Edge Sampling

每条 NURBS 曲线沿弧长方向采样 4 个三维点



C) Scalar Quantization

坐标归一化到 $[-1, 1]^3$ 并进行 4000 级均匀标量量化



词表设计 / Vocabulary Design

统一 token ID 空间: 几何坐标、局部面编号与控制符

- Face index tokens: 0-49
- Coordinate tokens: 50-4049
- SPECIAL tokens: START / SEP / END
- PAD only for training

1

Topology-aware Sequentialization

输入: 点块 + adjacency; 输出: 排序 / 重编号

DFS face sorting

MAX adjacent face-index edge sorting

re-index local face ids

原始顺序

重排

局部重编号

有序拓扑编号

2

Block Sequence Construction

START

face blocks

edge blocks

END

face_block = 49 tokens: 16 points $\times$ 3 coordinates + 1 face index

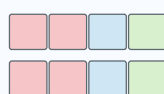
edge_block = 14 tokens: 2 adjacent face indices + 4 points $\times$ 3 coordinates

完整 token 序列

3

Autoregressive Transformer Generation

输入令牌序列



LLaMA3-style Decoder-only Transformer

RoPE

RMSNorm

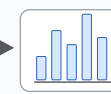
SwiGLU

GQA

$\times L$ layers

输出: next-token 概率, 采样得到生成序列

next-token



p(下一令牌)

生成序列

4

OpenCASCADE Reconstruction

dequantization

NURBS surface / curve fitting

Sewing / Solid construction

STEP export

BRepCheck validity evaluation

输出: STEP 文件 + BRepCheck 有效性报告

数据预处理修正

评估反馈 / 训练反馈

模型训练信号